

Krasikov–Lagarias certificates for the $3x + 1$ problem are tempered roulette measures

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Abstract

We report verified certificates for the Krasikov–Lagarias difference-inequality system for the $3x + 1$ problem at 3-adic depths $k = 13, 15, 17, 19, 20$, giving $\pi_1(x) \geq x^\gamma$ for all large x with $\gamma = 0.9146$ at $k = 20$ (the published record of the method is $\gamma = 0.84$). We then identify what these certificates *are*: to $R^2 > 0.99$, each Perron eigenvector is a pure power of a single exactly solvable measure — the stationary law (the *roulette measure*) of the geometric- w model on residue classes mod 3^j , for which we give the closed form $\pi(1, 2, 4, 5, 7, 8) = (8, 16, 11, 4, 2, 22)/63$ at $j = 2$. The tempering exponent α_k rises $0.8024 \rightarrow 0.8785$ across the five depths and an out-of-sample prediction at $k = 20$ was confirmed to four decimals. We prove an edge-rate theorem: implicit differentiation of the min-loss identity at $(\lambda, q) = (2, 1)$ gives $d\gamma/dq = 1/\log(4/3) = 3.4761$, and the measured ratios $(1 - \gamma_k)/(3.4761(1 - q_k)) = 0.824, 0.847, 0.873, 0.908$ converge to 1: the transfer from eigenvector homogenization to density exponent is analytic. A mass conservation lemma shows the heterogeneity cascade of the eigenvector sits exactly on the conservative line ($a + c \leq 1$) precisely because $2^{\log_2 3} = 3$, and the fine-end injection saturates: $\text{CV}_1(k) = 0.5136 - 0.337(0.910)^k$ to residual 10^{-6} over nine depths. The remaining open core of the full-density program $\gamma \rightarrow 1$ is a single drift-direction statement: $a < c$ uniformly. All claims are labeled proved, verified, measured, or conjectured.

1 Introduction

Let $T(n) = n/2$ for even n and $T(n) = (3n + 1)/2$ for odd n , and let $\pi_1(x)$ denote the number of integers $n \leq x$ whose T -orbit reaches 1. The $3x + 1$ conjecture asserts $\pi_1(x) = \lfloor x \rfloor$; see Lagarias [2] for the classical survey and Tao [3] for the strongest known metric result (almost all orbits attain almost bounded values). The strongest known *density* lower bounds come from the difference-inequality method of Krasikov and Lagarias [1], which produces, from a finite feasibility certificate at 3-adic depth k , a bound $\pi_1(x) \geq x^{\gamma_k}$ for all sufficiently large x . Krasikov and Lagarias obtained $\gamma = 0.84$.

This paper does three things.

1. **New certificates (Section 2).** We push the exact Krasikov–Lagarias system to depths $k = 13, 15, 17, 19, 20$ (up to $3^{19} = 1,162,261,467$ classes) and verify feasibility, obtaining

$$\gamma = 0.8624, 0.8805, 0.8953, 0.9069, 0.9146.$$

2. **Identification of the certificates (Sections 3 and 4).** The certificate eigenvectors are not opaque numerical objects. We exhibit an exactly solvable probability measure on the classes mod 3^j — the stationary measure of the geometric- w residue dynamics, which we call the *roulette measure* — and show empirically that at every computed depth the eigenvector is a pure power of it,

$$\text{eigvec} = \text{roulette}^{\alpha_k}, \quad R^2 = 0.9927 \dots 0.9977,$$

with tempering exponent $\alpha_k \uparrow$ and an out-of-sample prediction at $k = 20$ confirmed to four decimals (Conjecture T: $\alpha_k \rightarrow 1$).

3. **An analytic transfer and a proof program (Sections 5 and 6).** We prove that the density exponent responds to the homogenization parameter q at the exact linear rate $d\gamma/dq = 1/\log(4/3) = 3.4761$ at the edge $(\lambda, q) = (2, 1)$, so that $\alpha_k \rightarrow 1$ (equivalently $q_k \rightarrow 1$) would give $\gamma \rightarrow 1$: $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$. We then prove a mass conservation lemma pinning the eigenvector's heterogeneity cascade to the conservative line $a + c \leq 1$ — an identity that holds precisely because $2^{\log_2 3} = 3$, i.e., by the defining constant of the problem — and reduce the whole program to one measured-but-unproven inequality: the drift direction $a < c$, uniformly in k .

Throughout, every assertion carries one of four labels. *Proved*: complete mathematical proof. *Verified*: finite exact or floating-point computation, reproducible from stated parameters. *Measured*: statistical fit or extrapolation from computed data. *Conjectured*: neither proved nor finitely checkable. Section 7 collects the ledger.

Notation

$\beta := \log_2 3 = 1.58496 \dots$ (we reserve α for the tempering exponent). For odd n , the Syracuse map is $U(n) = (3n + 1)/2^{w(n)}$ with $w(n) = v_2(3n + 1)$. For a finite family of positive reals, CV denotes the coefficient of variation (standard deviation over mean). For $j \geq 1$, $[3^j] := \{m \bmod 3^j : m \equiv 2 \pmod{3}\}$, so $|[3^j]| = 3^{j-1}$.

2 The Krasikov–Lagarias system and certificates at depth 13–20

2.1 The inequality system

Fix $k \geq 2$ and $\lambda \in (1, 2]$. The nontrivial-tree Krasikov–Lagarias system $L_k(\lambda)$ [1] asks for a vector $c : [3^k] \rightarrow \mathbb{R}_{>0}$ satisfying, for every $m \in [3^k]$,

$$m \equiv 2 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-2} \bar{c}\left(\frac{4m-2}{3}\right), \quad (1)$$

$$m \equiv 5 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m), \quad (2)$$

$$m \equiv 8 \pmod{9} : \quad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-1} \bar{c}\left(\frac{2m-1}{3}\right), \quad (3)$$

where $4m$ is taken mod 3^k , the branch targets are taken mod 3^{k-1} , and for $t \in [3^{k-1}]$,

$$\bar{c}(t) := \min\{c(t), c(t + 3^{k-1}), c(t + 2 \cdot 3^{k-1})\} \quad (4)$$

is the minimum over the three refinements of t in $[3^k]$.

The provenance of the three edge types is the backward Syracuse tree. A tree node in class m receives preimages of size $\approx 4m/3$ (via $(4m-2)/3$, weight $(3/4)^\gamma = \lambda^{\beta-2}$ under the ansatz $\pi_m(x) \sim c(m)x^\gamma$ with $\lambda = 2^\gamma$), of size $\approx 2m/3$ (via $(2m-1)/3$, weight $(3/2)^\gamma = \lambda^{\beta-1}$), and of size $4m$ (two halvings, weight $4^{-\gamma} = \lambda^{-2}$); classes $m \equiv 5 \pmod 9$ have branch target divisible by 3 and lose it (multiples of 3 have no odd preimages). The minimum (4) is what makes a *finite* system sound: the branch target is only known mod 3^{k-1} , so the certificate may claim only the worst of its three refinements. Krasikov–Lagarias prove [1] that feasibility of $L_k(\lambda)$ implies

$$\pi_1(x) \geq x^\gamma, \quad \gamma = \log_2 \lambda,$$

for all sufficiently large x . Both class maps in the system are bijections: $m \mapsto 4m$ on $[3^k]$, and each branch map from its mod-9 type onto $[3^{k-1}]$ — a fact we use in Proposition 7.

The right-hand sides of (1)–(3) define a monotone, positively homogeneous (topical) operator F_λ ; feasibility is equivalent to its nonlinear Perron value being ≥ 1 , and the critical λ_k (the *edge*) is located by bisection, with the Perron eigenvector computed by power iteration.

2.2 The certificates

Theorem 1 (Certificates; verified). *The system $L_k(\lambda_k)$ is feasible at the values of Table 1. Consequently $\pi_1(x) \geq x^{0.9146}$ for all sufficiently large x .*

k	classes 3^{k-1}	λ_k	$\gamma_k = \log_2 \lambda_k$	q_k
13	531,441	1.8181	0.8624	0.9519
15	4,782,969	1.8411	0.8805	0.9594
17	43,046,721	1.8601	0.8953	0.9655
19	387,420,489	1.8751	0.9069	0.9705
20	1,162,261,467	1.8852	0.9146	0.9723

Table 1: Verified certificates. q_k is the min-loss parameter of Proposition 7. At depth $k = 11$ the same code reproduces $\gamma = 0.84$, the exponent proved in [1].

Verification is by direct substitution of the stored eigenvector into all 3^{k-1} inequalities at the stated λ_k (at $k = 20$: 1.16×10^9 constraints, chunked out-of-core, checkpointed power iteration to $\|Fc - c\|/\|c\| < 10^{-3}$ followed by a feasibility polish). The certificates and verification scripts are archived with the research dossier.

The numbers γ_k in Table 1 climb steadily but with visibly shrinking increments. The remainder of the paper is about *why*: the certificate eigenvectors have an exact structural description, and the residual distance $1 - \gamma_k$ is governed by a single homogenization parameter with an analytic exchange rate.

3 The roulette measure

3.1 The geometric- w model

At the level of natural density the Syracuse map is a solved stochastic object. The relevant exactness statement goes back to Terras [4]: each pattern of the first r steps of T occupies exactly one residue class mod 2^r . In Syracuse coordinates: for odd n , prescribing the first m cascade exponents $(w_1, \dots, w_m) = (j_1, \dots, j_m)$ confines n to exactly one residue class mod $2^{j_1 + \dots + j_m + 1}$,

so the relative density among odd integers of that cylinder is exactly $\prod_i 2^{-j_i}$. Thus, in density, the w_i are independent geometric($1/2$) variables: $P(w = j) = 2^{-j}$, $j \geq 1$. (The independence of w from the current residue class mod 3^j — cross-base blindness — is exact in the limit of large 2-adic modulus; measured leakage at finite modulus vanishes as the modulus grows. We label the model *density-exact in the 2-adic direction, measured in the cross-base direction*.)

Definition 2 (Geometric- w residue dynamics). Fix $j \geq 1$. The *geometric- w chain* on the classes mod 3^j coprime to 3 is

$$r_{t+1} = (3r_t + 1) \cdot 2^{-w_t} \pmod{3^j}, \quad w_t \text{ i.i.d., } P(w = m) = 2^{-m},$$

inverses taken in $(\mathbb{Z}/3^j)^\times$. Its stationary measure $\pi^{(j)}$ is the *roulette measure* at depth j .

The name: 2 is a primitive root mod 3^j for every j , so the multiplier 2^{-w} steps around the full cyclic group $(\mathbb{Z}/3^j)^\times$ (the wheel, of order $2 \cdot 3^{j-1}$) with geometrically decaying bets 2^{-w} ; the base point $3r + 1$ is re-spotted each turn. Two structural facts make the chain exactly solvable.

Proposition 3 (Hierarchical solvability; proved). (i) *The base map $r \mapsto 3r + 1 \pmod{3^j}$ depends only on $r \pmod{3^{j-1}}$. (ii) Consequently the stationary measure at depth j is the explicit linear push-forward of the stationary measure at depth $j - 1$:*

$$\pi^{(j)}(s) = \sum_{\rho \pmod{3^{j-1}}} \pi^{(j-1)}(\rho) \mu_j(s \cdot (3\rho + 1)^{-1}),$$

where $\mu_j(u)$ is the total geometric weight of $\{w \geq 1 : 2^{-w} \equiv u \pmod{3^j}\}$, namely $\mu_j(2^{-m}) = 2^{-m}/(1 - 2^{-\text{ord}_j(2)})$ for $1 \leq m \leq \text{ord}_j(2) = 2 \cdot 3^{j-1}$. In particular $\pi^{(j)}$ is computable in closed form by downward recursion, with no eigenproblem.

Proof. (i) is immediate: $3(r + d3^{j-1}) + 1 = (3r + 1) + d3^j \equiv 3r + 1$. For (ii): the one-step law from r depends only on $\rho = r \pmod{3^{j-1}}$ by (i), so stationarity of $\pi^{(j)}$, whose projection to depth $j - 1$ is the stationary $\pi^{(j-1)}$ of the projected chain, reads exactly as displayed. The weights μ_j aggregate the geometric series over the residue of w modulo the order of 2. $\square$

3.2 Closed form at depth two

Theorem 4 (Roulette measure mod 9; proved). *The stationary measure of the geometric- w chain on $\{1, 2, 4, 5, 7, 8\} \pmod{9}$ is*

$$\pi(1, 2, 4, 5, 7, 8) = \frac{(8, 16, 11, 4, 2, 22)}{63}.$$

Proof. Marginal mod 3. Since $3r + 1 \equiv 1 \pmod{3}$ and $2^{-w} \equiv 2^w \pmod{3}$, the next state is $\equiv 2 \pmod{3}$ iff w is odd, with probability $\sum_{w \text{ odd}} 2^{-w} = 2/3$, independent of r . Hence $\pi(r \equiv 1) = 1/3$, $\pi(r \equiv 2) = 2/3$.

Wheel mod 9. $\text{ord}_9(2) = 6$ and the inverse powers $2^{-w} \pmod{9}$ for $w \equiv 1, \dots, 6 \pmod{6}$ are 5, 7, 8, 4, 2, 1, with aggregated weights $\mu(5, 7, 8, 4, 2, 1) = (32, 16, 8, 4, 2, 1)/63$ by Proposition 3(ii) with $1 - 2^{-6} = 63/64$.

Base mod 9. By Proposition 3(i) the base depends only on $r \pmod{3}$: if $r \equiv 1$ then $3r + 1 \equiv 4 \pmod{9}$; if $r \equiv 2$ then $3r + 1 \equiv 7 \pmod{9}$.

Assembly. Multiplying the wheel by the two bases:

$$4 \cdot (5, 7, 8, 4, 2, 1) \equiv (2, 1, 5, 7, 8, 4), \quad 7 \cdot (5, 7, 8, 4, 2, 1) \equiv (8, 4, 2, 1, 5, 7) \pmod{9}.$$

Hence, weighting the two rows by $1/3$ and $2/3$,

$$\pi(s) = \frac{1}{3} \mu(4^{-1}s) + \frac{2}{3} \mu(7^{-1}s),$$

which evaluates to $\pi(1) = (16 + 2 \cdot 4)/189 = 8/63$, $\pi(2) = (32 + 2 \cdot 8)/189 = 16/63$, $\pi(4) = (1 + 2 \cdot 16)/189 = 11/63$, $\pi(5) = (8 + 2 \cdot 2)/189 = 4/63$, $\pi(7) = (4 + 2 \cdot 1)/189 = 2/63$, $\pi(8) = (2 + 2 \cdot 32)/189 = 22/63$. These sum to 1 and reproduce the marginals $\pi(1) + \pi(4) + \pi(7) = 1/3$; stationarity is a direct check. $\square$

The wheel is heavily biased: $\pi(8)/\pi(7) = 11$. This bias is not noise — it is the exact fingerprint of the geometric halving law wrapped around the cyclic group generated by 2, and (Section 4) it is the shape the Krasikov–Lagarias certificates are made of. Numerically, the closed form matches the empirical residue distribution of Syracuse orbits mod $9/27/81$ with correlations $0.997/0.997/0.992$ (measured).

Remark 5 (Why the roulette governs the backward tree). The linearization of the edge system $L_k(2)$ obtained by replacing the minimum (4) by the triple mean is, after transposition, the geometric- w transition operator. Computed exactly at $j = 4, 5, 6$ ($27/81/243$ classes), its spectrum is: leading eigenvalue 1 with eigenvector the roulette measure, and *all* other eigenvalues of modulus exactly $1/4 = \lambda^{-2}$ (verified). Linear damping toward the roulette is thus overwhelming; whatever heterogeneity survives in the true (min-coupled) eigenvector must be re-injected at each level by the min-nonlinearity. This is the mechanistic frame for everything that follows.

4 The tempering law

We now compare the certificate eigenvectors of Section 2 with the roulette measure. For each depth k , aggregate the eigenvector into the 729 blocks $m \bmod 3^7$ ($m \equiv 2 \bmod 3$) and regress log block-mass against log roulette block-mass (the roulette measure conditioned to classes $\equiv 2 \bmod 3$ and renormalized).

Conjecture 6 (T: Tempering law; measured at five depths). *The Krasikov–Lagarias certificate is a tempered roulette measure:*

$$\text{eigvec}_k = \text{roulette}^{\alpha_k} \quad (\text{pure power law, no further structure}), \quad \alpha_k \longrightarrow 1.$$

k	α_k	R^2	$1 - \alpha_k$
13	0.8024	0.9927	0.1976
15	0.8291	0.9949	0.1709
17	0.8509	0.9963	0.1491
19	0.8682	0.9973	0.1318
20	0.8785	0.9977	0.1215

Table 2: The tempering law at block depth mod 3^7 (measured). The fit is a single-parameter power law; R^2 increases with depth.

Three measured facts accompany Table 2.

1. **Residual identification.** Define $\text{CV}_{\text{res}}(k)$ as the coefficient of variation of the multiplicative residual eigvec/roulette over the blocks. Then numerically $1 - \alpha_k = \text{CV}_{\text{res}}(k)$ (e.g. 0.1976 vs 0.197 at $k = 13$; agreement within 0.003 at all measured depths). Tempering exponent and residual heterogeneity are the same number seen twice.
2. **The grand chain.** The measured ratios $(1 - \gamma_k)/\text{CV}_{\text{res}}(k) = 0.6991, 0.6973, 0.6971, 0.6985$ are constant to three decimals, giving the empirical law

$$\gamma_k = 1 - 0.698(1 - \alpha_k).$$

Section 5 derives the analytic skeleton of the constant 0.698.

3. **Out-of-sample confirmation.** Before the $k = 20$ certificate finished, the extrapolations of Tables 2 and 3 were registered as predictions: α_{20} predicted 0.878, measured 0.8785; cascade root θ_{20} predicted 0.849, measured 0.8488. The tempering law also passed a referee test one depth earlier: the residual amplitude at $k = 19$ was predicted at 0.131 from $k \leq 17$ data and measured at 0.1333 with shape correlation 0.9991.

Conjecture T is a statement about a family of finite, exactly defined nonlinear eigenproblems; each instance is finitely checkable, and five instances have been checked. What is conjectural is the limit $\alpha_k \rightarrow 1$.

5 The analytic transfer: the edge-rate theorem

Why should homogenization ($\alpha_k \rightarrow 1$) yield full density ($\gamma_k \rightarrow 1$)? The bridge is an exact identity satisfied by every edge eigenvector, plus an implicit-function computation at the corner $(\lambda, q) = (2, 1)$.

Proposition 7 (Min-loss identity; proved). *Let λ_k be the critical value of L_k and c the associated Perron eigenvector, so that (1)–(3) hold with equality. Put $S = \sum_{m \in [3^k]} c(m)$ and $S_{\min} = \sum_{t \in [3^{k-1}]} \bar{c}(t)$. Then, exactly,*

$$1 = \lambda_k^{-2} + \frac{q_k}{3} \left(\lambda_k^{\beta-2} + \lambda_k^{\beta-1} \right), \quad q_k := \frac{3 S_{\min}}{S} \in (0, 1]. \quad (5)$$

Moreover $q_k = 1$ iff every refinement triple of c is constant, and $q = 1$ forces $\lambda = 2$, i.e. $\gamma = 1$.

Proof. Sum the edge equalities over all $m \in [3^k]$. The map $m \mapsto 4m$ is a bijection of $[3^k]$, so the doubling terms sum to $\lambda^{-2}S$. The branch map $m \mapsto (4m - 2)/3$ is a bijection from $\{m \equiv 2 \pmod 9\}$ onto $[3^{k-1}]$, and $m \mapsto (2m - 1)/3$ is a bijection from $\{m \equiv 8 \pmod 9\}$ onto $[3^{k-1}]$; hence each branch family contributes its weight times $S_{\min}$, giving $S = \lambda^{-2}S + (\lambda^{\beta-2} + \lambda^{\beta-1})S_{\min}$, which is (5). Since $\bar{c}(t)$ is the minimum of a triple whose mean summed over t is $S/3$, we get $q_k \leq 1$ with equality iff all triples are flat. Finally, at $q = 1$ the right side of (5), $h(\lambda) = \lambda^{-2} + \frac{1}{3}(\lambda^{\beta-2} + \lambda^{\beta-1})$, is strictly decreasing on $(1, 2]$ (elementary calculus) and $h(2) = \frac{1}{4} + \frac{1}{3}(\frac{3}{4} + \frac{3}{2}) = 1$, so $\lambda = 2$ is the unique solution. $\square$

Identity (5) was additionally verified to eight decimals on the computed eigenvectors for $k = 5, \dots, 11$. It says: *the entire gap between the Krasikov–Lagarias method and full density is the min-loss $1 - q_k$, the price of knowing branch targets only mod 3^{k-1} .*

Theorem 8 (Edge rate; proved). *Along the curve (5), with $\gamma = \log_2 \lambda$,*

$$\left. \frac{d\gamma}{dq} \right|_{(\lambda,q)=(2,1)} = \frac{1}{\log(4/3)} = 3.4761\dots$$

Consequently, to first order at the edge, $1 - \gamma \sim 3.4761(1 - q)$.

Proof. Write $F(\lambda, q) = \lambda^{-2} + \frac{q}{3}(\lambda^{\beta-2} + \lambda^{\beta-1}) - 1$. At $(2, 1)$:

$$F_q = \frac{1}{3} \left(2^{\beta-2} + 2^{\beta-1} \right) = \frac{1}{3} \left(\frac{3}{4} + \frac{3}{2} \right) = \frac{3}{4}, \quad F_\lambda = -\frac{2}{8} + \frac{1}{3} \left((\beta-2)\frac{3}{8} + (\beta-1)\frac{3}{4} \right) = \frac{3(\beta-2)}{8},$$

using $2^{\beta-3} = 3/8$ and $2^{\beta-2} = 3/4$. Hence

$$\frac{d\lambda}{dq} = -\frac{F_q}{F_\lambda} = \frac{2}{2-\beta}, \quad \frac{d\gamma}{dq} = \frac{1}{\lambda \log 2} \cdot \frac{d\lambda}{dq} = \frac{1}{(2-\beta) \log 2} = \frac{1}{\log 4 - \log 3} = \frac{1}{\log(4/3)}. \quad \square$$

Corollary 9 (First-order exactness at the edge; measured). *The measured ratios*

$$\frac{1 - \gamma_k}{3.4761(1 - q_k)} = 0.824, 0.847, 0.873, 0.908 \quad (k = 13, 15, 17, 19)$$

increase monotonically toward 1: the finite- k data approach the tangent line of Theorem 8. In particular the empirical constant 0.698 of Section 4 is a finite- k composite, $0.698 = 3.4761 \times (1 - q_k)/\text{CV}_{\text{res}}(k)$, and the transfer from homogenization to density exponent is analytic: the only open content of the program $\gamma \rightarrow 1$ is $q_k \rightarrow 1$.

6 The proof program: from $\alpha \rightarrow 1$ to $\gamma \rightarrow 1$

By Proposition 7 and Theorem 8, the chain of implications is

$$\alpha_k \rightarrow 1 \iff \text{CV}_{\text{res}}(k) \rightarrow 0 \iff q_k \rightarrow 1 \xrightarrow{\text{Thm 8}} \gamma_k \rightarrow 1 \implies \pi_1(x) \geq x^{1-\varepsilon} \quad \forall \varepsilon > 0. \quad (6)$$

(The first two equivalences are the measured identities of Section 4 together with the min–mean linearization: for certificate triples, $E[1 - \min/\text{mean}] = c_1 \text{CV} + c_2 \text{CV}^2$ with $c_1 \in [1.19, 1.45]$, so $1 - q_k = \Theta(\text{CV}_{\text{top}}(k))$; measured.) It remains to control the heterogeneity of the eigenvector. Define, for each trit position p of the class label, $\text{CV}_p(k)$ as the mean CV over sibling triples of classes differing only in trit p ; $p = 1$ is the fine end, $p = k - 2$ the top. The min-loss parameter is controlled by the top: $1 - q_k \approx 1.36 \text{CV}_{\text{top}}(k)$ (measured). The difference field of the eigenvector propagates between adjacent trit scales: a pure trit offset $\delta = d \cdot 3^P$ satisfies $4\delta = \delta + 3\delta$ under the doubling map (scales $P, P+1$) and $4\delta/3 = \delta/3 + \delta$ under the branch maps (scales $P-1, P$), giving an exact pointwise transport identity for differences of c (verified: correlation 1.000000, mean residual 2.8×10^{-4} on 15,115 samples at $k = 13$). Averaging over classes yields the two-coefficient closure

$$\text{CV}_P = a \text{CV}_{P-1} + c \text{CV}_{P+1} \quad (\text{fit error} \leq 1.2\% \text{ at } k = 13, 15, 17, 19; \text{measured}). \quad (7)$$

6.1 Subcriticality is forced by the defining constant

Lemma 10 (Mass conservation at the edge; proved). *At $\lambda = 2$ set $W_0 = \lambda^{-2} = \frac{1}{4}$, $W_2 = \lambda^{\beta-2} = \frac{3}{4}$, $W_8 = \lambda^{\beta-1} = \frac{3}{2}$. The row masses of the three constraint types (1)–(3) are*

$$W_0 + W_2 = 1 \quad (m \equiv 2 \bmod 9), \quad W_0 = \frac{1}{4} \quad (m \equiv 5), \quad W_0 + W_8 = \frac{7}{4} \quad (m \equiv 8),$$

and their average is exactly 1:

$$3W_0 + W_2 + W_8 = \frac{3}{4} + 3 \cdot 2^{\beta-2} = \frac{3}{4} + \frac{9}{4} = 3, \quad \text{precisely because } 2^\beta = 3.$$

Proof. Direct evaluation; the last equality is $3 \cdot 2^{\beta-2} = 3 \cdot \frac{2^\beta}{4} = \frac{9}{4}$ iff $2^\beta = 3$. $\square$

Corollary 11 (Subcriticality of the cascade; proved within the closure (7)). *Each unit of difference-field mass at trit scale P is redistributed to scales $\{P-1, P, P+1\}$ with total weight equal to the row mass of its class (Lemma 10, via the offset algebra above); the minimum in (4) is 1-Lipschitz, so contributes no expansion. Averaging and applying the triangle inequality bounds the closure coefficients of (7), after absorbing the self-term, by*

$$a + c \leq 1.$$

Measured at $k = 20$: $a + c = 0.9955$, the strictness being the measured incoherence factor ≈ 0.90 between the two transport channels.

Corollary 11 is the conceptual center of the program: the identity $2^\beta = 3$ — the defining equation of the $3x + 1$ problem — is precisely what pins the average row mass to 1. The heterogeneity cascade of the certificate can therefore *never* grow exponentially: the system sits exactly on the conservative line, and the only question is the *direction of drift* along it.

6.2 The source is bounded: fine-end saturation

Proposition 12 (Fine-end saturation; measured at nine depths). *The finest-level triple CV of the certificate (each depth at its own critical λ_k) satisfies*

$$\text{CV}_1(k) = 0.5136 - 0.337(0.910)^k$$

with residual 10^{-6} across nine depths in $8 \leq k \leq 20$: the injection at the fine end converges geometrically to the finite limit ≈ 0.514 . The mid-cascade damping ratio $\text{CV}_{P+1}/\text{CV}_P$ is measured ≤ 0.86 uniformly over the interior of the profile.

(At $k = 20$ the formula predicts $\text{CV}_1 = 0.4625$; measured 0.4629.) Note the single rate 0.910 recurring in three roles — fine-end saturation, per-depth homogenization, and the decay of $1 - q_k$ — one mechanism, three faces (measured).

6.3 The one remaining statement

On the conservative line $a + c = 1$ with $c > a$, the decaying solution of (7) is $\text{CV}_P \sim \theta^P$ with root $\theta = a/c < 1$: heterogeneity injected at the fine end dies geometrically before reaching the top trit, whence $\text{CV}_{\text{top}}(k) \lesssim \text{CV}_1 \cdot \theta^k$ and the chain (6) closes. The measured root is comfortably below 1 and converging (Table 3).

k	a	c	$a + c$	$\theta = a/c$
13				0.8248
15				0.8360
17				0.8438
19				0.8480
20	0.4706	0.5249	0.9955	0.8488

Table 3: Lattice coefficients (measured). The θ increments shrink by ≈ 0.6 per depth step; extrapolated limit $\theta_\infty \approx 0.85$. The $k = 20$ row was predicted out-of-sample (θ predicted 0.849, measured 0.8488).

Open Problem 13 (Drift direction). *Prove that the closure coefficients of (7) satisfy $a < c$ uniformly in k ; equivalently, that $\theta_k = a_k/c_k \leq \bar{\theta} < 1$ for all k . Together with Lemma 10/Corollary 11 (no exponential growth), Proposition 12 (bounded source), and Theorem 8 (analytic transfer), this yields $\gamma_k \rightarrow 1$, i.e. $\pi_1(x) \geq x^{1-\varepsilon}$ for every $\varepsilon > 0$.*

Two remarks on the status of this reduction. First, the closure (7) is itself a measured (mean-field) step: the exact transport identity is pointwise and verified, but its averaged two-coefficient form carries a $\leq 1.2\%$ fit error; deriving (a, c) from the weight structure of the exact identity is part of the open problem. Second, even the full conclusion $\gamma \rightarrow 1$ is the *density* statement $\pi_1(x) \geq x^{1-\varepsilon}$, the ceiling of the Krasikov–Lagarias method; it is not the $3x+1$ conjecture, which requires every integer, not almost every integer in density exponent.

7 Honest status and outlook

Proved	Min-loss identity (5) (Prop. 7); edge rate $d\gamma/dq = 1/\log(4/3)$ (Thm. 8); roulette solvability and closed form mod 9 (Prop. 3, Thm. 4); mass conservation $3W_0 + W_2 + W_8 = 3$ (Lem. 10); feasibility $\Rightarrow$ density bound (Krasikov–Lagarias [1]).
Verified (finite computation)	Certificates $k = 13, 15, 17, 19, 20$ with $\gamma = 0.8624/0.8805/0.8953/0.9069/0.9146$ (Thm. 1); identity (5) to 8 decimals at $k = 5 \dots 11$; linear edge spectrum $\{1\} \cup \{ \mu = 1/4\}$ at $j = 4, 5, 6$ (Rem. 5); exact pointwise transport identity (residual 2.8×10^{-4}); geometric- w cylinder densities exact (Terras [4]).
Measured	Tempering law α_k , R^2 (Table 2); $1 - \alpha_k = \text{CV}_{\text{res}}(k)$; constant 0.698; edge-rate ratios $0.824 \rightarrow 0.908$ (Cor. 9); closure (7) and coefficients (Table 3); $a + c = 0.9955$ at $k = 20$; fine-end saturation (Prop. 12); out-of-sample hits at $k = 19, 20$.
Conjectured	Conjecture T: $\alpha_k \rightarrow 1$; Open Problem 13: $a < c$ uniformly ($\theta_\infty < 1$).

The picture that emerges is simple to state. The Krasikov–Lagarias certificate — the best known rigorous handle on the density of $3x+1$ convergence — is not an unstructured numerical artifact: it is, to $R^2 > 0.99$ at a billion classes, one exactly solvable measure raised to one scalar power. The distance of that power from 1 is the distance of the density exponent from 1, through an exchange rate that is an explicit derivative, $1/\log(4/3)$. The defining constant of

the problem pins the certificate’s heterogeneity cascade to the conservative line; the source of heterogeneity saturates; damping is spectrally overwhelming at the linear level. What is not yet proved is one inequality about the direction of drift along the conservative line.

The framework is falsifiable at the next depth: at $k = 21$ it predicts $\gamma \approx 0.918$, $\theta \approx 0.850$, and a tempering fit with $R^2 > 0.9977$.

Reproducibility

All certificates, verification logs, and the scripts generating every number in this paper are archived in the research dossier accompanying this work (directories `certificates/`, `scripts/`, `findings/`).

Acknowledgment of method

The computations, fits, and drafts were produced in an extended human–AI collaboration; every proved statement above has a self-contained proof in the text, independent of that provenance.

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